Design and Implementation of a Fire Fighting System Using Microcontroller

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Abstract

This study aims to design and implementation a fire fighting system using Atmega32 microcontroller. This system accomplished by connecting Atmega32 microcontroller to temperature sensors, push buttons, Liquid Crystal Display (LCD) and other components. The system monitors the sensors values continuously to warn or inform the user in abnormal situations or at fire danger. This warning takes different ways such as the appearance of the threat type on the system screen or the operation of different alarm sirens or by sending a Short Message Service (SMS), at some cases the system sends command to fire fighting unit to extinguish fire. The system also has many options that achieve more functions.

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Chapter One

Introduction

1.1 General Concept

Fire accident is common feature in factories, houses, markets etc. in every country. Due to poor fire protection, lack of adequate fire alarm and emergency exit, fire increases death. With the advancement of human civilization, fire-safety has been a prime concern. Fire hazards can be fatal and denigrating for industrial and household security, also minatory for human life.

The best way to reduce these losses is to respond to the emergency situation as quick as possible. So, there comes the necessity of standalone autonomous fire detection systems. These systems render the works of quick detection, alarm notification, and sometimes initiation of fire extinguishing. A fire or smoke alarm system can be monitored locally in the premises, or remotely at a distant place as prerequirement. Remote alarm system provides the owner of the premise with the advantage of monitoring from distant location and taking immediate actions when an emergency message is received, unlike a manual system. Remote monitoring systems can be designed in various waysusing wireless sensor networks, Ethernet, image processing and other digital communication technologies.

Fire, smoke or the leak of gases threat people's life and their properties. Most building and facilities in Sudan free from fire alarm systems. Firefighting system is used to prevent, extinguish, localize, or block fires in enclosed spaces. Automatic fire-fighting system, which is very important to protect people and properties, is installed in buildings and rooms where the fire hazard is comparatively high.

1.2 Statement of Problem

Although The Existing Fire fighting Systems are reliable and have a wide range of pros, they are accompanied by concerns about being complex, incompact, non-standalone, expensive and having redundant appurtenances.

Therefore, there is necessity for a system which would be reliable and swift responsive as well as simple, easy implementable and cost effective.

1.3 Objectives

The main objectives of this study are to:

1. Design a control circuit for automatic fire fighting system.
2. Simulate the proposed automatic fire fighting system control circuit.
3. Implement the proposed automatic fire fighting system control circuit.

1.4 Methodology

This study develops a microcontroller based system to detect smoke and heat at a house or facilities, by analyzing this signal and take specific action. The system can operate at auto or manual mode in which the system acts as detector only and the owner can activate the actuators remotely as needed.

In this study the system is divided into parts; hardware and software. The hardware a sensor circuitry is designed to develop system awareness and capability to detect over-temperature, smoke and flame. Microcontroller Atmega16 is used to process the various sensor signals and control the system actuators accordingly. A software code is developed to control the overall system functions. The code is written in BASIC language using basic compiler for AVR (BASCOM-AVR). A Global System Mobile (GSM) module is interconnected to the system and used to send SMS indicating the system and environment status. System software simulation is done using Proteus package.

1.5 Layout

This research consists of five chapters, Chapter one contains general introduction, defines problem, objectives and methodology. Chapter two contains a general background and general information to the main parts used in the study. Chapter three handles the hardware and software considerations. Chapter four deals with the system implementation and testing. Finally, chapter five presents the conclusion and recommendations.

Chapter Two

Theoretical Background

2.1 Introduction

All fire alarm systems essentially operate on the same principle. If a detector detects smoke or heat or someone operates a break glass unit (manual break point), then alarm sounders operate to warn others in the building that there may be a fire and to evacuate. When choosing a fire alarm system many factors need to be considered such as the purpose, building structure and current legislation.

Fire safety in new, extended or altered buildings is the responsibility of the local authority who delegates the responsibility to building control department who enforces and administers the building regulations. In existing commercial buildings, premises are subject to the regulatory reform (fire safety) order 2005. The order states that the 'responsible person' within the organization has a legal duty to undertake a suitable and sufficient fire risk assessment at their premises. This assessment is a detailed report of the fire risk in your building, the adequacy of existing fire precautions and the need for any additional fire precautions. The fire risk Assessment will determine the level of fire alarm system required for business premises. The guidance usually indicates appropriate British standards. The main standard for fire alarm systems is BS5839 pt1:2013[1].

2.2 Sensors

A sensor is generally defined as an input device that provides a usable output in response to a specific physical quantity input. The physical quantity input that is to be measured, called the measurand, affects the sensor in a way that causes a response represented in the output. The output of many modern sensors is an electrical signal, but alternatively, could be a motion, pressure, flow, or other usable type of output. Some examples of sensors include a thermocouple pair, which converts a temperature difference into an electrical

output; a pressure sensing diaphragm, which converts a fluid pressure into a force or position change; and a Linear Variable Differential Transformer (LVDT), which converts a position into an electrical output [2].

The devices that inform the control system about what is actually occurring are called sensors (also known as transducers). As an example, the human body has an amazing sensor system that continually presents our brain with a reasonably complete picture of the environment whether we need it all or not. For a control system, the designer must ascertain exactly what parameters need to be monitored. For example, position, temperature, and pressure and then specify the sensors and data interface circuitry to do the job[3].

Classification of sensors is conventionally by the convention principle, the quantity being measured, the technology used, or the application [4]. Most sensors work by converting some physical parameter such as temperature or position into an electrical signal. This is why sensors are also called transducers, which are devices that convert energy from one form to another. The various types of sensors are the position sensors, angular velocity sensors, proximity sensors, load sensors, pressure sensors, flow Sensors, liquid-level sensors, vision sensors, temperature sensors, etc [3].

2.2.1 Smart and intelligent sensors

Modern definition of smart or intelligent sensors can be formulated now by the following way: Smart sensor is an electronic device, including sensing element, interfacing, signal processing and one or several intelligence functions as self-testing, self-identification, self-validation or self-adaptation. The key word in this definition is intelligence. The self-adaptation is relatively new function of smart sensors. Novel designed self-adaptation smart sensor systems are based on so-called adaptive algorithms, which were used at the first time in various digital measuring systems [5].

Strong growth expected for sensors based on Microelectromechanical system technologies (MEMS), smart sensors and sensors with bus capabilities.

Smart sensors' capability to have more intelligence built into them continues to drive their application in automotive, aerospace and defense, industrial, medical, and most recently – homeland security applications. Proprietary algorithms customized for specific applications analyze sensor data on key parameters to optimize machining, processing, and other component product or process quality [5]. Intelligent wireless sensor-based controls have drawn industry attention on account of reduced costs, better power management, ease in maintenance, and effortless deployment in remote and hard-to-reach areas. They have been successfully deployed in many industrial applications such as maintenance, monitoring, control, security, etc [6].

Because of semiconductor technologies and its Moore's law along with the MEMS technology, sensors are becoming more cost effective, smaller, less power hungry, with more embedded features. The sensors not only provide digital data but can also provide the expected information. For instance, functions like portrait/landscape, specific tap and free fall detections can be left to the sensor due to its capability to detect such simple information with values preset by the device manufacturer [7].

2.2.2 Temperature sensors

Temperature is defined as a specific degree of hotness or coldness as referenced to a specific scale. It can also be defined as the amount of heat energy in an object or system. Heat energy is directly related to molecular energy (vibration, friction and oscillation of particles within a molecule): the higher the heat energy, the greater the molecular energy. Temperature sensors detect a change in a physical parameter such as resistance or output voltage that corresponds to a temperature change [8].

Temperature sensors give an output proportional to temperature. Most temperature sensors have a positive temperature coefficient (desirable), which means that the sensor output goes up as the temperature goes up, but some sensors have a negative temperature coefficient, which means that the output goes down as the temperature goes up. Many control systems require

temperature sensors, if only to know how much to compensate other sensors that are temperature-dependent. Some common types are discussed next.

- Bimetallic temperature sensors: The bimetallic temperature sensor consists of a bimetallic strip wound into a spiral. The bimetallic strip is a laminate of two metals with different coefficients of thermal expansion. As the temperature rises, the metal on the inside expands more than the metal on the outside, and the spiral tends to straighten out. These sensors are typically used for on-off control as in a household thermostat where a mercury switch is rocked from on to off.

- Thermocouples: The thermocouple was developed over 100 years ago and still enjoys wide use, particularly in high-temperature situations. The thermocouple is based on the See beck effect, a phenomenon whereby a voltage that is proportional to temperature can be produced from a circuit consisting of two dissimilar metal wires.

- Resistance temperature detectors: The Resistance Temperature Detector (RTD) is a temperature sensor based on the fact that metals increase in resistance as temperature rises. RTDs have the advantage of being very accurate and stable (characteristics do not change over time). The disadvantages are low sensitivity (small change in resistance per degree), relatively slow response time to temperature changes, and high cost.

- Thermistors: A thermistor is a two-terminal device that changes resistance with temperature. Thermistors are made of oxide-based semiconductor materials and come in a variety of sizes and shapes. Thermistors are nonlinear; therefore, they are not usually used to get an accurate temperature reading but to indicate temperature changes, for example, overheating. Also, most thermistors have a negative temperature coefficient, which means the resistance decreases as temperature increases.

- Integrated-circuit temperature sensors: Integrated circuit (IC) temperature sensors come in various configurations. A common example is the

LM34 and LM35 series. The LM34 produces an output voltage that is proportional to Fahrenheit temperature, and the LM35 produces an output that is proportional to Celsius temperature. Notice that it has three active terminals: supply voltage (Vs), ground, and Vout. The output voltage of the LM35 is directly proportional to °C, that is, Vout = 10mV/°C This equation states that for each 1° increase in temperature, the output voltage increases by 10mV. If only positive temperatures need to be measured, then the simple circuit shown in the spec sheet (bottom middle of datasheet) can be used. If positive and negative temperatures must be measured, then the circuit on the bottom right can be used, which requires a positive and negative supply voltage. The LM35 is a convenient IC to work with because the output voltage is in degrees Celsius. Some ICs, such as the LM135, provide an output that is in degrees kelvin. One degree of kelvin or Celsius represents the same interval of temperature, but the Kelvin scale starts at absolute zero temperature, which is 273°C below freezing. There is also an absolute zero temperature scale for Fahrenheit degrees, called the Rankine scale. **2.2.3 Smoke detectors**

Smoke detectors operate on two main principles: ionization detectors which will detect the ionized air from a fire even when there is little or no visible smoke, and the optical type, which is design to detect smoke which is present even when there is little or no rise in temperature. In general, the ionization types are used industrially and the optical types in domestic uses. The ionization type of detector uses a radioactive source, usually americium241, with a low activity level, typically 0.8μCi [4].

The gap between the source plate and the electrode is normally conducting due to the emission of alpha particles (helium ions) by the americium. The ion current is very small, of the order of 10pA, so that any additional insulation leakage would be substantial in comparison. The acceptable level of insulation leakage is of the order of 0.5pA. This implies that the insulators must not be touched, and if the ionization chamber has to be replaced, utmost care must be taken to avoid any contamination of the

insulation by, for example, solder flux. In the presence of smoke from a fire, particles entering the ionization chamber will be struck by the alpha particles, and the alpha particles will cling to the much larger particles of smoke. Because the charged units are now much larger, they cannot travel so quickly in air, and the current is reduced. By detecting the reduction in current, the detector can be made to activate an alarm.

➢ Ionization smoke sensor: Ionization smoke sensor contains a small amount of radioactive material, americium embedded in a gold foil matrix within an ionization chamber. The matrix is made by rolling gold and americium oxide ingots together to form a foil approximately one micrometer thick. This thin gold-americium foil is then sandwiched between a thicker (~0.25 millimeter) silver backing and a 2 micron thick palladium laminate. This is thick enough to completely retain the radioactive material, but thin enough to allow the alpha particles to pass.

The ionization chamber as shown in Figure 2.1 is basically two metal plates a small distance apart. One of the plates carries a positive charge, the other a negative charge. Between the two plates, air molecules-made up mostly of oxygen and nitrogen atoms-are ionized when electrons are kicked out of the molecules by alpha particles from the radioactive material (alpha particles are big and heavy compared to electrons). The result is oxygen and nitrogen atoms that are positively charged because they are short one electron; the free electrons are negatively charged.

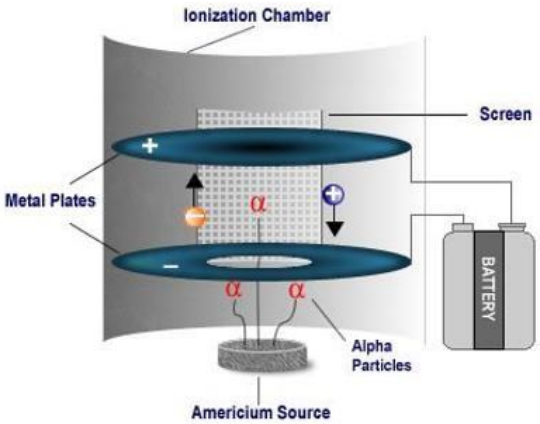


Figure 2.1: The ionization chamber

The Figure 2.2 illustrates how ionization technology works. The positive atoms flow toward the negative plate, as the negative electrons flow toward the positive plate. The movement of the electrons registers as a small but steady flow of current. When smoke enters the ionization chamber, the current is disrupted as the smoke particles attach to the charged ions and restore them to a neutral electrical state. This reduces the flow of electricity between the two plates in the ionization chamber. When the electric current drops below a certain threshold, the sensor sends a signal telling there is a smoke.

In the smoke-free chamber, positive and negative ions create a small current as they migrate to charged plates. Smoke particles and combustion gases interact with the ions generated by the alpha particles, restoring them to their neutral electronic state and decreasing the electrical current passing through the cell Figure 2.2. As fewer ions are available to migrate to the plates, the disrupted current causes the smoke sensor triggers the microcontroller [9].

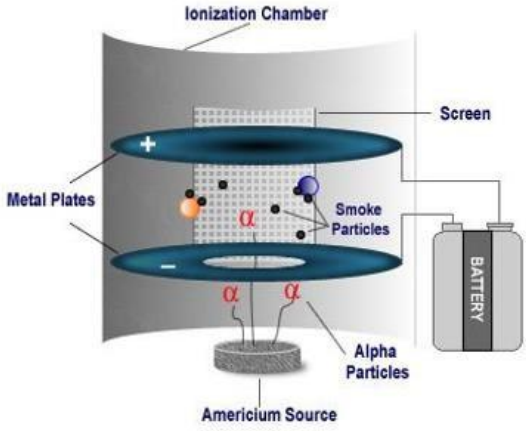


Figure 2.2: Smoke detector inside ionization chamber

2.2.4 Infrared Sensor

All objects emit infrared energy provided their temperature is above absolute zero (0 Kelvin). There is a direct correlation between the infrared energy an object emits and its temperature. Infrared Sensor (IR) measure the infrared energy emitted from an object in the 4–20 micron wavelength and convert the reading to a voltage. Typical IR technology uses a lens to concentrate radiated energy onto a thermopile. The resulting voltage output is

amplified and conditioned to provide a temperature reading. Factors that affect the accuracy of IR sensing are the reflectivity (the measure of a material's ability to reflect infrared energy), transitivity (the measure of a material's ability to transmit or pass infrared energy), and emissivity (the ratio of the energy radiated by an object to the energy radiated by a perfect radiator of the surface being measured). An object that has an emissivity of 0.0 is a perfect reflector, while an object with an emissivity of 1.0 emits (or absorbs) 100% of the infrared energy applied to it [8].

2.3 Microprocessor

The digital IC called a microprocessor, has ushered in a whole new era for control systems electronics. This revolution has occurred because the microprocessor brings the flexibility of program control and the computational power of a computer to bear on any problem. Automatic control applications are particularly well suited to take advantage of this technology, and microprocessor-based control systems are rapidly replacing many older control systems based on analog circuits or electromechanical relays. One of the first microprocessor-based controllers made specifically for control applications was the Programmable Logic Controller (PLC). A microprocessor by itself is not a computer; additional components such as memory and input/output circuits are required to make it operational. However, the microcontroller, which is a close relative of the microprocessor, does contain all the computer functions on a single IC. Microcontrollers lack some of the power and speed of the newer microprocessors, but their compactness is ideal for many control applications; most so-called microprocessor-controlled devices, such as vending machines, are really using microcontrollers. Some specific reasons for using a digital, microprocessor design in control systems are the following:

- Low-level signals from sensors, once converted to digital, can be transmitted long distances virtually error-free.
- A microprocessor can easily handle complex calculations and control strategies.

- Long-term memory is available to keep track of parameters in slowmoving systems.

Changing the control strategy is easy by loading in a new program; no hardware changes are required. Microprocessor-based controllers are more easily connected to the computer network within an organization. This allows designers to enter program changes and read current system status from their desk terminals. To perform all microprocessor functions, the microprocessor incorporates various functional units in an appropriate manner. Such an internal structure or organizational structure of microprocessor, which determines how it operates, is known as its architecture. Typical microprocessor architecture is shown in Figure 2.3.

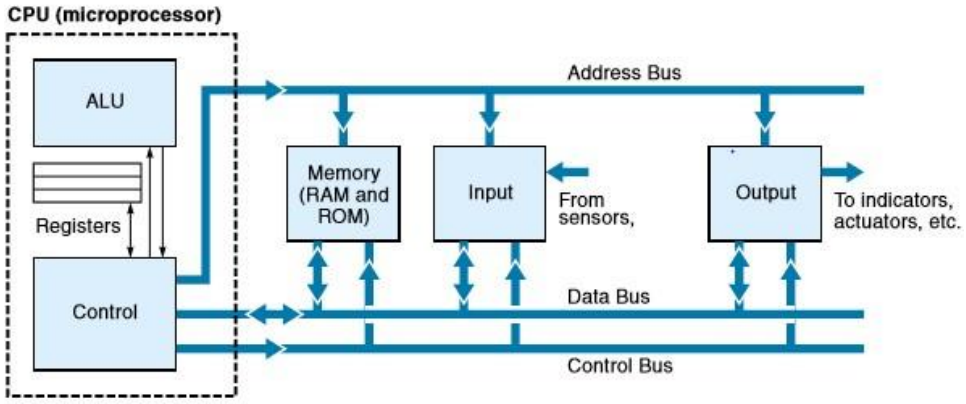


Figure 2.3: Architecture of Microprocessor

2.3.1 Busses

Microcomputer, like all computers, manipulates binary information. The binary information is represented by binary digits, called bits. Microprocessor operates on a group of bits which are referred to as a word. The number of bits making- microprocessor a word varies with the microprocessor. Common word sizes are 4, 8, 12 and 16 bits (microprocessors with 32 bit-word have also of late entered the market). Other binary terms that will be of interest in subsequent discussions are the byte and the nibble, which represent a set of 8 bits and 4 bits, respectively. Figure 2.3 shows busses interconnecting various blocks. These busses allow exchange of words between the blocks. A bus has a wire or line for each bit and thus allows exchange of all bits of a word in

parallel. The processing of bits in the microprocessor is also in parallel. The busses can thus be viewed as data highways. The width of a bus is the number of signal lines that constitute the bus.

The figure shows for simplicity three busses for distinct functions. Over the address bus, the microprocessor transmits the address of that Input Output (I/O) device or memory locations which it desires to access. This address is received by all the devices connected to the processor, but only the device which has been addressed responds. The data bus is used by the microprocessor to send and receive data to and from different devices (I/O and memory) including instructions stored in memory. Obviously the address bus is unidirectional and the data bus is bi-directional. The control bus is used for transmitting and receiving control signals between the microprocessor and various devices in the system.

2.3.2 Arithmetic Logic Unit

The Arithmetic Logic Unit (ALU) is a combinational network that performs arithmetic and logical operations on the data.

2.3.3 Internal registers

A number of registers are normally included in the microprocessor. These are used for temporary storage of data, instructions and addresses during execution of a program. Those in the Intel microprocessor 8085 microprocessor are typical and are described below:

(i) Accumulator (Acc) or result register: This is an 8-bit register used in various arithmetic and logical operations. Out of the two operands to be operated upon, one comes from accumulator (Acc), whilst the other one may be in another internal register or may be brought in by the data bus from the main memory. Upon completion of the arithmetic/logical operation, the result is placed in the accumulator (replacing the earlier operand). Because of the later function, this register is also called as result register.

(ii) General purpose registers or scratch pad memory: There are six general purpose 8-bit registers that can be used by the programmer for a variety of

purposes. These registers, labelled as B, C, D, E, H and L, can be used individually (e.g.,when operation on 8-bit data is desired) or in pairs (e.g., when a 16-bit address is to be stored). Only B-C, D-E and H-L pairs are allowed.

(iii) Instruction Register (IR): This 8-bit register stores the next instruction to be executed. At the proper time this stored word (instruction) is fed to an instruction decoder which decodes it and supplied appropriate signals to the control unit. When the execution has been accomplished the new word in the instruction register is processed.

(iv) Program Counter (PC): This is a 16-bit register which holds the address of the next instruction that has to be fetched from the main memory and loaded into the instruction register. The program controlling the operation is stored in the main memory and instructions are retrieved from this memory normally in order. Therefore, normally the address contained in the PC is incremented after each instruction is fetched. However, certain classes of instruction can modify the PC so that the programmer can provide for branching away from the normal program flow. Examples are instructions in the "jump" and 'call subroutine' groups.

(v) Stack Pointer (SP): This is also a 16-bit register and is used by the programmer to maintain a stack in the memory while using subroutines.

(vi) Status register or condition flags: A status register consisting of a few flip-flops, called as condition flags (in 8085 the number of flags is five) is used to provide indication of certain conditions that arise during arithmetic and logical operations. These are:

'zero' Flag is set if result of instruction is 0.

'sign' Set if Most Significant Bit (MSB) of result is 1.

'parity' Set if result has even parity.

'carry' Set if carry or borrow resulted.

'auxiliary carry' Set if instruction caused a carry out of bit 3 and into bit 4 of the resulting value.

(vii) Dedicated registers: Several other registers are incorporated in the microprocessor for its internal operation. They cannot be accessed by the programmer and hence do not concern much a microprocessor user.

2.3.4 Instruction decoder and control unit

It decodes each instruction and under the supervision of a clock controls the external and internal units ensuring correct logical operation of the system[10].

2.4 Microcontroller

A microcontroller is a single-chip computer. Micro suggests that the device is small, and controller suggests that it is used in control applications. Another term for microcontroller is embedded controller, since most of the microcontrollers are built into or embedded in the devices they control [11]. A microcontroller is a computer with most of the necessary support chips onboard. All computers have several things in common, namely:

1- A Central Processing Unit (CPU) that 'executes' programs.
2- Some Random Access Memory (RAM) where it can store data that is variable.
3- Some Read Only Memory (ROM) where programs to be executed can be stored.
4- I/O devices that enable communication to be established with the outside world i.e. connection to devices such as keyboard, mouse, monitors and other peripherals.

There are a number of other common characteristics that define microcontrollers. If a computer matches a majority of these characteristics, then it can be classified as a 'microcontroller'.

2.4.1 Microcontroller features

Microcontrollers from different manufacturers have different architectures and different capabilities. Some may suit to a particular

application while some others may be totally unsuitable. The hardware features of microcontrollers in general are described below.

- Supply voltage: Most microcontrollers operate with the standard 5V supply. Some microcontrollers can operate at as low as 2.7V and some will tolerate 6V without any problems. the allowed limits of the supply voltage in the datasheet.
- The clock: All microcontrollers require an oscillator (known as a clock) to operate. Most microcontrollers will operate with a crystal and two capacitors. Some will operate with resonators or with external resistor-capacitor pair. Some microcontrollers have built-in resistor-capacitor type oscillators and they do not require any external timing components (e.g. PIC12C672). Resonators are not as stable as the crystals but they are more stable than the resistorcapacitor networks. Crystal oscillators should be chosen for applications which require very accurate timing. For applications where the timing stability requirements are very modest, resonators should be chosen. If the application is not time sensitive you should consider using external or internal (if available) resistor-capacitor timing components for simplicity and low cost.
- Timers: Timers are an important part of any microcontroller. A timer is basically a counter which is driven from an accurate clock (or a division of this clock). Timers can be 8-bits or 16-bits long. Data can be loaded into the timers and they can be started and stopped under software control. Most timers can be configured to generate an interrupt when they reach a certain count (usually when they overflow).
- Watchdog: Many microcontrollers have at least one watchdog facility, also known as the Watch Dog Timer (or WDT). A WDT is usually an 8-bit timer with a prescaler option and is clocked from a free running on-chip oscillator. The watchdog is usually refreshed by the user program at regular intervals and a reset occurs ifthe program fails to refresh the watchdog. Watchdog facilities are commonly used in real-time systems where it is required to check the proper

termination of one or more activities. All PIC microcontrollers are equipped with a WDT.

- Interrupts: Interrupts are a very important concept in microcontrollers. An interrupt causes a microcontroller to respond to external and internal (e.g. timer) events very quickly. When an interrupt occurs the microcontroller leaves its normal flow of execution and jumps directly to the Interrupt Service Routine (ISR). The source of an interrupt can either be internal or external. Internal interrupts are usually generated by the built-in timer circuits when the timer count reaches a certain value. External interrupts are generated by the devices connected external to the microcontroller and these interrupts are asynchronous, i.e. it is not known when an external interrupt will be generated. An example is the Analogue-to-Digital (A/D) conversion complete interrupt, which is generated when a conversion is completed.

- Analogue-to-digital converter: Some microcontrollers are equipped with A/D converter circuits. Usually these converters are 8-bits, but some microcontrollers have 10- or even 12-bit converters. Some microcontrollers have multiple A/D channels (e.g. PIC16F877 is equipped with eight A/D channels). A/D converters usually generate interrupts when a conversion is complete so that the user program can read the converted data very quickly. A/D converters are very useful in control and monitoring applications since most sensors produce analogue output voltages.

- Serial input-output: Some microcontrollers contain hardware to implement serial asynchronous communications interface. The baud rate and the data format can usually be selected in software by the programmer. The built-in timer circuits are usually used to generate an accurate baud rate. If serial I/O hardware is not provided, it is easy to develop software to implement serial data transfer using any I/O pin of a microcontroller. Some microcontrollers incorporate Serial Peripheral Interface (SPI), Integrated Interconnect (I_2C), or Controller Area Network (CAN) bus interfaces. These enable a microcontroller to interface to other compatible devices easily, usually over a bus structure.

- In-circuit programming: In microcontroller development cycle, a microcontroller is normally removed from its socket and then programmed using a programmer device. The programmed chip is then re-inserted into its socket, ready for testing. This is usually very tedious work, especially during the development of complex software projects. In-circuit programming enables a microcontroller to be programmed while the chip is in the applications circuit, i.e. there is no need to remove the chip for programming. This feature speeds up the program development cycle considerably.
- Electrically Erasable Programmable ROM (EEPROM): Data memory EEPROM type memory is also very common in many microcontrollers. The programmer can store non-volatile data in such memory and can also change this data whenever required. For example, if the microcontroller is used to measure the temperature, the maximum and minimum values during a period can be stored in an EEPROM type memory. Some microcontroller types provide between 64 and 256 bytes of EEPROM data memories, while some others do not have any such memories.
- Pulse Width Modulation (PWM) output: Some microcontrollers provide PWM outputs which can be used in some electronic applications. One such application is to provide an effective analog output from a microcontroller by varying the duty cycle of the PWM output. It is possible to modify the period or the duty cycle of a PWM output by loading the appropriate registers [12].

2.4.2 Microcontroller architectures

Basically, two types of architectures are used in microcontrollers: Von Neumann architecture and Harvard architecture. Von Neumann architecture is used by a very large percentage of microcontrollers and here all memory space is on the same bus and instruction and data are treated identically. In the Harvard architecture (used by the PIC microcontrollers), code and data storage are on separate buses and this allows code and data to be fetched simultaneously, resulting in a more efficient implementation.

2.4.3 Microcontroller types

The predominant families of microcontrollers are 8-bit types since this word size has proved popular for the vast majority of tasks the devices have been required to perform. The single byte word is regarded as sufficient for most purposes and has the advantage of easily interfacing with the variety of IC memories and logic circuitry currently available. The serial acronym for the American Standard Code for Information Interchange (ASCII) data is also byte sized making data communications easily compatible with the microcontroller devices. Because the type of application for the microcontroller may vary enormously most manufacturers provide a family of devices, each member of the family capable of fitting neatly into the manufacturer's requirements. This avoids the use of a common device for all applications where some elements of the device would not be used; such a device would be complex and hence expensive.

The microcontroller family would have a common instruction subset but family members differ in the amount, and type, of memory, timer facility, port options, etc. possessed, thus producing cost-effective devices suitable for particular manufacturing requirements. Memory expansion is possible with offchip RAM and/or ROM; for some family members there is no on-chip ROM, or the ROM is either Electrically Programmable ROM (EPROM) or EEPROM known as flash EEPROM which allows for the program to be erased and rewritten many times. Additional on-chip facilities could include Analogue to Digital Conversion (ADC), Digital-to-Analogue Conversion (DAC) and analogue comparators. Some family members include versions with lower pin count for more basic applications to minimize costs [12].

2.4.4 Comparison between microcontroller and microprocessor

Microcontrollers are general purpose microprocessors which have additional parts that allow them to control external devices. Basically, a microcontroller executes a user program which is loaded in its program memory. Under the control of this program, data is received from external

devices (inputs), manipulated and then data is sent to external output devices. A microcontroller is a very powerful tool that allows a designer to create sophisticated I/O data manipulation algorithms.

Microcontrollers are classified by the number of bits in a data word. 8bit microcontrollers are the most popular ones and are used in many applications. 16-bit and 32-bit microcontrollers are much more powerful, but usually more expensive and not required in many small to medium general purpose applications where microcontrollers are used. The simplest microcontroller architecture consists of a microprocessor, memory, and I/O. The microprocessor consists of a central CPU and the Control Unit (CU). The CPU is the brain of a microprocessor and is where all of the arithmetic and logical operations are performed. The control unit controls the internal operations of the microprocessor and sends out control signals to other parts of the microprocessor to carry out the required instructions[13].

A microcontroller may take an input from the device it is controlling and controls the device by sending signals to different components in the device. A microcontroller is often small and low cost. The components may be chosen to minimize size and to be as inexpensive as possible. The actual processor used to implement a microcontroller can vary widely. In many products, such as microwave ovens, the demand on the CPU is fairly low and price is an important consideration. In these cases, manufacturers turn to dedicated microcontroller chips – devices that were originally designed to be low-cost, small, low-power, embedded CPUs [13]. A microprocessor differs from a microcontroller in a number of ways. The main distinction is that a microprocessor requires several other components for its operation, such as program memory and data memory, input-output devices, and an external clock circuit. A microcontroller, on the other hand, has all the support chips incorporated inside its single chip. All microcontrollers operate on a set of instructions (or the user program) stored in their memory. A microcontroller fetches the instructions from its program memory one by one, decodes these

instructions, and then carries out the required operations. Microcontrollers have traditionally been programmed using the assembly language of the target device. Although the assembly language is fast, it has several disadvantages. An assembly program consists of mnemonics, which makes learning and maintaining a program written using the assembly language difficult. Also, microcontrollers manufactured by different firms have different assembly languages, so the user must learn a new language with every new microcontroller he or she uses. Microcontrollers can also be programmed using a high-level language, such as BASIC, PASCAL, or C. High-level languages are much easier to learn than assembly languages.

In theory, a single chip is sufficient to have a running microcontroller system. In practical applications, however, additional components may be required so the microcomputer can interface with its environment. With the advent of the PIC family of microcontrollers the development time of an electronic project has been reduced to several hours. Basically, a microcomputer executes a user program which is loaded in its program memory. Under the control of this program, data is received from external devices (inputs), manipulated, and then sent to external devices (outputs). A microcontroller is a very powerful tool that allows a designer to create sophisticated input-output data manipulation under program control. Microcontrollers are classified by the number of bits they process. Microcontrollers with 8-bits are the most popular and are used in most microcontroller-based applications. Microcontrollers with 16 and 32 bits are much more powerful, but are usually more expensive and not required in most small- or medium-size general purpose applications that call for microcontrollers. Memory, an important part of a microcontroller system, can be classified into two types: program memory and data memory. Program memory stores the program written by the programmer and is usually nonvolatile (i.e., data is not lost after the power is turned off). Data memory stores the temporary data used in a program and is usually volatile [11].

2.5 Buzzer

For alarm purposes a lot of electric bells, alarms and buzzers are available in the markets that have got different prices and uses. The buzzer that will use in this thesis is a 5-12V buzzer and has got enough alarm sound to be used in a fire alarm system. Louder buzzer would have been even better but then their operating voltages are high as we had a supply of maximum up to 12V available with us on the board.

2.6 GSM System

A GSM modem is a wireless modem that works with a GSM wireless network. A wireless modem behaves like a dial-up modem. The main difference between them is that a dial-up modem sends and receives data through a fixed telephone line while a wireless modem sends and receives data through radio waves. Like a GSM mobile phone, a GSM modem requires a SIM card from a wireless carrier in order to operate. GSM modem provides full functional capability to serial devices to send SMS and data over GSM network. The product is available as board level or enclosed in metal box. The board level product can be integrated in to various serial devices in providing them SMS and data capability and the unit housed in a metal enclosure can be kept outside to provide serial port connection. The GSM modem supports popular "AT" command set so that users can develop applications quickly. The product has SIM card holder to which activated SIM card is inserted for normal use. The power to this unit can be given from UPS to provide uninterrupted operation. This product provides great feasibility for devices in remote location to stay connected which otherwise would not have been possible where telephone lines do not exist. Application areas are:

- Mobile transport vehicles.
- LAN based SMS servers
- Alarm notification of critical events including servers
- Network monitoring and SMS reporting
- Data transfer applications from remote locations

- Monitor and control of serial services through GSM Network
- Integration to custom software for warehouse, stock, production, dispatch notification through SMS, automatic meter reading, ect.…

2.6.1 SMS communications

Short message service is a type of communications process that enables the transmission of short text messages and data transfers to and from mobile devices such as cell phones. Messages are usually limited from 140 to 160 characters in length and are stored and forwarded at SMS centers. This allows messages and data transfers to be retrieved immediately or at a later time via an SMS center.

2.6.2 Benefits of using SMS

SMS communications provide an affordable and convenient means to send and receive data using mobile devices such as cell phones. Businesses and industry often require 24-hour coverage of their operations and have personnel who are on-call after normal work hours to handle work-related issues and emergencies. There are employees who are responsible for the proper functioning of equipment and processes at remote sites. Managers need to be notified of significant events.

Chapter Three

System Hardware and Software Considerations

3.1 System Description

The fire fighting system consists of: Detector group which can be a smoke, flame or heat detector, input keys that allow the user to do different actions, microcontroller Atmega 32, display, a group of LEDs, ULN2003A driver, and the output relays.

3.2 System Hardware Considerations

The main hardware components used to build this system are shown in Figure 3.1.

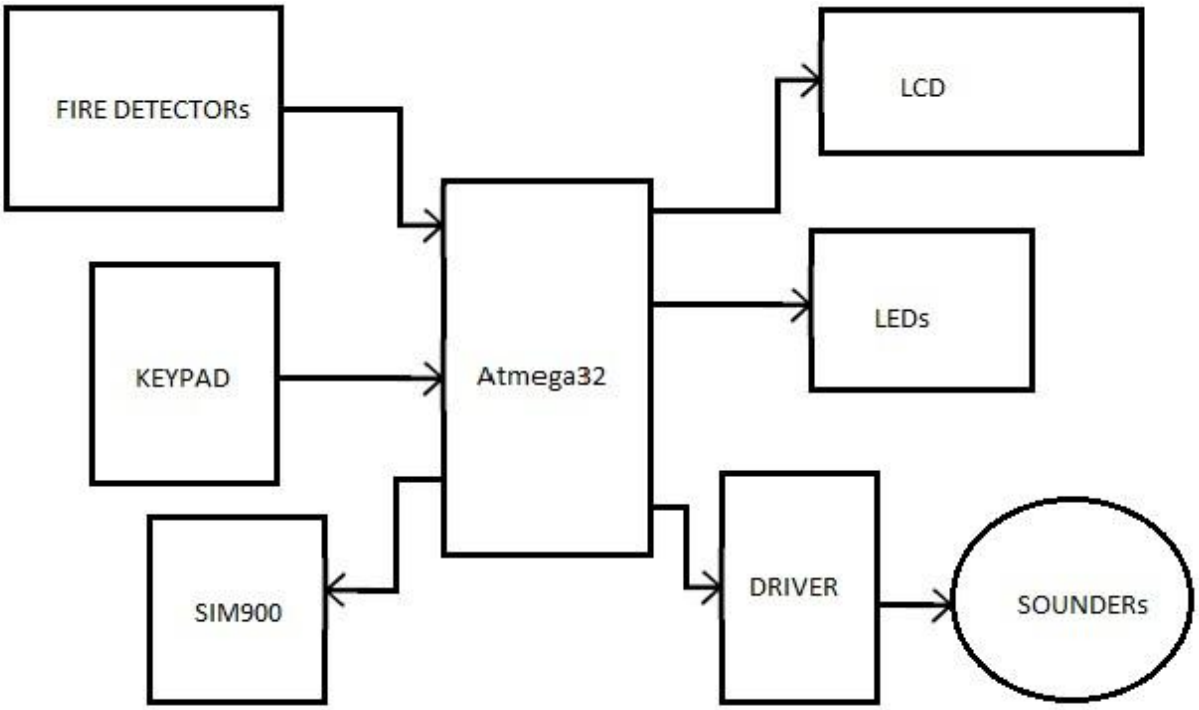


Figure 3.1: Block diagram for the circuit

3.2.1 Atmega32

Atmega32 microcontroller is the main brain of the study and it is hold the program of the operation conditions to be executed to reach the target. Atmga32 used because it is compatible to the study from the side of number of inputs/outputs, the internal memory to download the program and the other features, Figure 3.2 shows the microcontroller's pins.

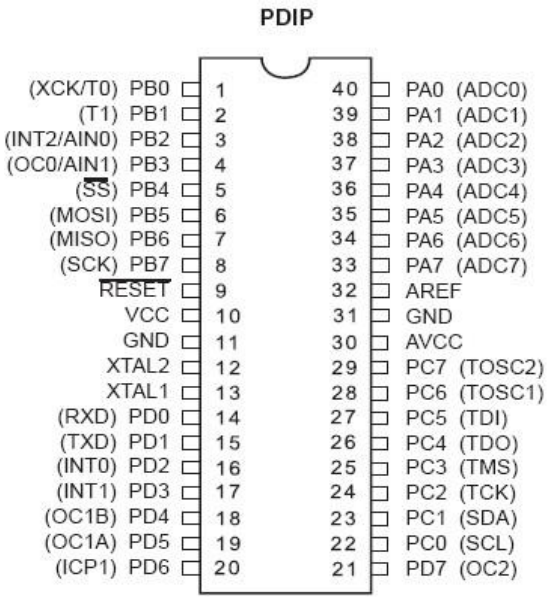


Figure 3.2: Atmega32

➢ Atmega32 features

• High-performance, low-power Atmel® AVR® 8-bit microcontroller

• Advanced RISC architecture

– 131Powerful instructions – Most single-clock cycle execution

– 32 x 8 general purpose working registers

– Fully static operation

– Up to 16MIPS throughput at 16MHz

– On-chip 2-cycle multiplier

• High endurance non-volatile memory segments

– 32Kbytes of in-system self-programmable flash memory

– 1024bytes EEPROM

– 2Kbyte internal SRAM

– Write/Erase cycles: 10,000 Flash/100,000 EEPROM

– Data retention: 20 years at 85°C/100 years at 25°C (1) – Optional boot code section with independent lock bits

In-system programming by on-chip boot program

True read-while-write operation

– Programming lock for software security

• JTAG (IEEE std. 1149.1 compliant) interface

– Boundary-scan capabilities according to the JTAG standard

– Extensive on-chip debug support

– Programming of flash, EEPROM, fuses, and lock bits through the JTAG interface.

• Peripheral features

– Two 8-bit timer/counters with separate prescalers and compare modes.

– One 16-bit timer/counter with separate prescaler, compare mode, and capture mode.

– Real time counter with separate oscillator.

– Four PWM channels.

– 8-channel, 10-bit ADC:

8 single-ended channels.

7 differential channels in TQFP package only.

2 differential channels with programmable gain at 1x, 10x, or 200x.

– Byte-oriented two-wire serial interface.

– Programmable serial USART.

– Master/Slave SPI serial interface.

– Programmable watchdog timer, separate on-chip oscillator.

– On-chip analog comparator.

• Special microcontroller features:

– Power-on reset and programmable brown-out detection.

– Internal calibrated RC oscillator.

– External and internal interrupt sources.

– Six sleep modes: Idle, ADC noise reduction, power save, power down, standby and extended standby.

• I/O and packages:

– 32 programmable I/O lines.

– 40-pin PDIP, 44-lead TQFP, and 44-pad QFN/MLF.

• Operating voltages:

– 2.7V-5.5V for ATmega32L.

– 4.5V-5.5V for ATmega32.

• Speed grades:

– 0-8MHz for ATmega32L.

– 0-16MHz for ATmega32.

• Power consumption at 1MHz, 3V, 25°C

–Active: 1.1mA

–Idle mode: 0.35mA

–Power-down mode: < 1μA ➢ Pin descriptions:

- VCC**:** Digital supply voltage.
- GND: Ground.
- Port A (PA7..PA0): Port A serves as the analog inputs to the A/D converter. Port A also serves as an 8-bit bi-directional I/O port, if the A/D converter is not used. Port pins can provide internal pull-up resistors (selected for each bit). The port A output buffers have symmetrical drive characteristics with both high sink and source capability. When pins PA0 to PA7 are used as inputs and are externally pulled low, they will source current if the internal pullup resistors are activated. The port A pins are tri-stated when a reset condition becomes active, even if the clock is not running. In the current system we config a part from port A as analog inputs (PA.0 to PA.3) to receive the signals from the different four zoon's. The internal A/D converts the received signal to binary system and divided with a suitable number to achieve the real value of the signal. PA.4 to PA.7 used as digital output.
- Port B (PB7..PB0): Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The port B output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, port B pins that are externally pulled low will

source current if the pullup resistors are activated. The port B pins are tri-stated when a reset condition becomes active, even if the clock is not running. PB.0 to PB.6 used as a digital input and receives to input keys status, port B.7 configured as digital output to Light Emitting Diodes (LED).

- Port C (PC7..PC0): Port C is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The port C output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, port C pins that are externally pulled low will source current if the pullup resistors are activated. The port C pins are tri-stated when a reset condition becomes active, even if the clock is not running. If the JTAG interface is enabled, the pull-up resistors on pins PC5(TDI), PC3(TMS) and PC2(TCK) will be activated even if a reset occurs. The TD0 pin is tri-stated unless TAP states that shift out data are entered. This port is used with the LCD pins PC.4 to PC.7 to transfer data to the LCD, PC.2 and PC.3 as enable and reset signals at LCD. Port C.0 and port C.1 are configured as digital outputs to LEDs.
- Port D (PD7..PD0): Port D is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The port D output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, port D pins that are externally pulled low will source current if the pullup resistors are activated. The port D pins are tri-stated when a reset condition becomes active, even if the clock is not running. PD.1 is used as a serial transmitter to send data to GSM module, The rest of this port is configured as a digital outputs to LEDs that preview the zoon's status two LEDs for each zoon represent the fault and alarm status.
- RESET: A low level on this pin for longer than the minimum pulse length will generate a reset, even if the clock is not running. Shorter pulses are not guaranteed to generate a reset.

- XTAL1: Input to the inverting oscillator amplifier and input to the internal clock operating circuit.
- XTAL2: Output from the inverting oscillator amplifier.
- AVCC: AVCC is the supply voltage pin for port A and the A/D converter. It should be externally connected to VCC, even if the A/D converter is not used. If the A/D converter is used, it should be connected to VCC through a low-pass filter.
- AREF**:** AREF is the analog reference pin for the A/D converter.

3.2.2 SIM900

SIM900 offers the benefit of sending short message to the user inform him with any change at his building, The positive edge of the signal is used to obtained this function that any change from zero to one at zoon's alarm or zoon's faults signals will send automatically in a short massage, Also if two pair of zoon's activate the counter of the actuator and the actuator status.

The SIM900 is a complete quad-band GSM/General Packet Radio Service (GPRS) solution in a Surface-Mount Device (SMT) module which can be embedded in the customer applications. Featuring an industry-standard interface, the SIM900 delivers GSM/GPRS 850/900/1800/1900MHz performance for voice, SMS, data, and fax in a small form factor and with low power consumption. With a tiny configuration of 24mmx24mmx3mm. SIM900 can fit almost all the space requirements in your M2M application, especially for slim and compact demand of design. Figure 3.3 shows the SIM900. The general features of SIM900 are:

- Quad-Band 850/ 900/ 1800/ 1900MHz.
- GPRS multi-slot class 10/8.
- GPRS mobile station class B.
- Compliant to GSM phase 2/2+
  Class 4 (2W @850/ 900MHz).

Class 1 (1W @ 1800/1900MHz).

- Dimensions: 24*24*3mm.
- Weight: 3.4g.
- Control via AT commands (GSM 07.07, 07.05 and SIMCOM enhanced AT Commands).
- SIM application toolkit.
- Supply voltage range 3.4 ... 4.5V.
- Low power consumption.
- Operation temperature: -30°C to +80°C.



Figure 3.3: SIM900

3.2.3 Fire detectors

There are many type of fire detector according to the detect principle that mentioned in section two, A heat detector is chosen because of the availability, simple, inexpensive and can detect wide range of temperature.

➢ LM35 Sensor

The LM35 series are precision integrated-circuit temperature sensors, with an output voltage linearly proportional to the centigrade temperature. Thus the LM35 has an advantage over linear temperature sensors calibrated in Kelvin, as the user is not required to subtract a large constant voltage from the output to obtain convenient centigrade scaling. The LM35 does not require any external calibration or trimming to provide typical accuracies of ±¼°C at room temperature and ±¾°C over a full -55°C to +150°C temperature range. The low

trimming and calibration at the wafer level. The low output impedance, linear output, and precise inherent calibration of the LM35 make interfacing to readout or control circuitry especially easy. The device is used with single power supplies, or with plus and minus supplies. As the LM35 draws only 60μA from the supply, it has very low self-heating of less than 0.1°C in still air. The LM35 is rated to operate over a −55°C to +150°C temperature range, while the LM35C is rated for a −40°C to +110°C range (−10° with improved accuracy). The LM35 series is available packaged in hermetic TO transistor packages, while the LM35C, LM35CA, and LM35D are also available in the plastic TO-92 transistor package. The LM35D is also available in an 8-lead surface-mount small outline package and a plastic TO-220 package. The main features of LM35 are:

- Calibrated directly in ° Celsius (Centigrade).
- Linear + 10 mV/°C scale factor .
- 0.5°C ensured accuracy (at +25°C).
- Rated for full −55°C to +150°C range. - Suitable for remote applications.
- Low cost due to wafer-level trimming.
- Operates from 4 to 30V.
- Less than 60μA current drain.
- Low self-heating, 0.08°C in still air.
- Nonlinearity only ±¼°C typical.
- Low impedance output, 0.1Ω for 1mA Load.

3.2.4 Liquid crystal display

LCD screen is an electronic display module and find a wide range of applications. A 16x2 LCD is a very basic module and is very common in various devices and circuits. They are preferred over seven segment displays. There are many advantages when compared to seven segment displays. They are: LCDs can display characters, numbers and even graphics [12]. A 16x2 LCD means it can display 16 characters per line and there are two such lines. In this LCD each

character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, command and data.

The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD. The reasons being: LCDs are economical; easily programmable; have no limitation of displaying special and even custom characters (unlike in seven segments), animations and so on.

Gnd: Ground pin 0VDC.
VCC: Supply voltage (4.7-5.3) VDC.
VEE: Variable resiste conect to this pin to adjust the contract.
RS: Selects command register when low, and data register when high.
R/W: Low to read from the register and high to write to the register.
EN: Sends data to data pins when a high to low pulse is given.
DB0-DB7 : Data pins used to transtfare data to LCD.uln
LED +: Backlight supply 5vdc LED
-: Backlight ground.

3.2.5 ULN2003 motor driver:

The ULN2003 is an array of seven NPN Darlington transistors capable of 500mA, 50V output. It features common-cathode flyback diodes for switching inductive loads. A Darlington transistor achieves very high current amplification by connecting two bipolar transistors in direct current (DC) coupling so the current amplified by the first transistor is amplified further by the second one. The resultant current gain is the product of those of the two component transistors. The seven Darlington pairs in ULN2003 can operate independently except the common cathode diodes that connect to their respective collectors. The main specifications of ULN2003 motor driver are:

- 500mA rated collector current (single output).
- 0V output (there is a version that supports 100V output).
- Includes output flyback diodes.
- Inputs compatible with TTL and 5-V CMOS logic.

The ULN2003 is known for its high-current, high-voltage capacity, as shown in Figure 3.5. The drivers can be paralleled for even higher current output. Even further, stacking one chip on top of another, both electrically and physically, has been done. Generally it can be used for interfacing with a stepper motor, where the motor requires high ratings which cannot be provided by other interfacing devices.

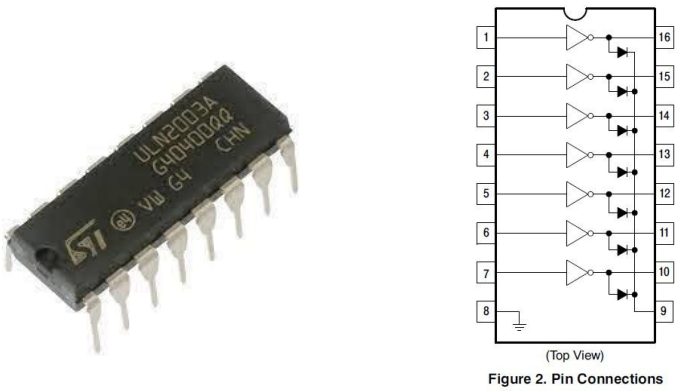


Figure 3.5: ULN2003 motor driver

3.2.6 Push buttons

A push button is a simple switch mechanism which permits user generated changes in the state of a circuit. Push button usually comes with four legs, legs are always connected in groups of two. When the pushbutton is pressed all the four legs are connected. Figure 3.6 shows the push button.

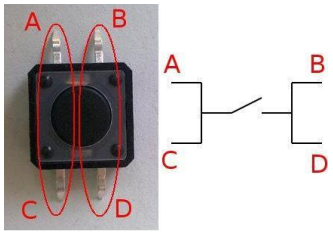


Figure 3.6: Push buttons

3.3 System Software Consideration

BASCOM language is used to write the software program and that language used because it is simple with enough capabilities to achieve the all condition and cases required. The Proteus program used to simulate the performance. **3.3.1 System code**

The system program is written using BASCOM language is shown in Appendix.

3.3.2 System simulation

The system is build up in the Proteus program as shown in Figure 3.7. The overall circuit shown in Figure 3.7 contains the input bush bottoms, sensor inputs LM35 (temperature sensors),

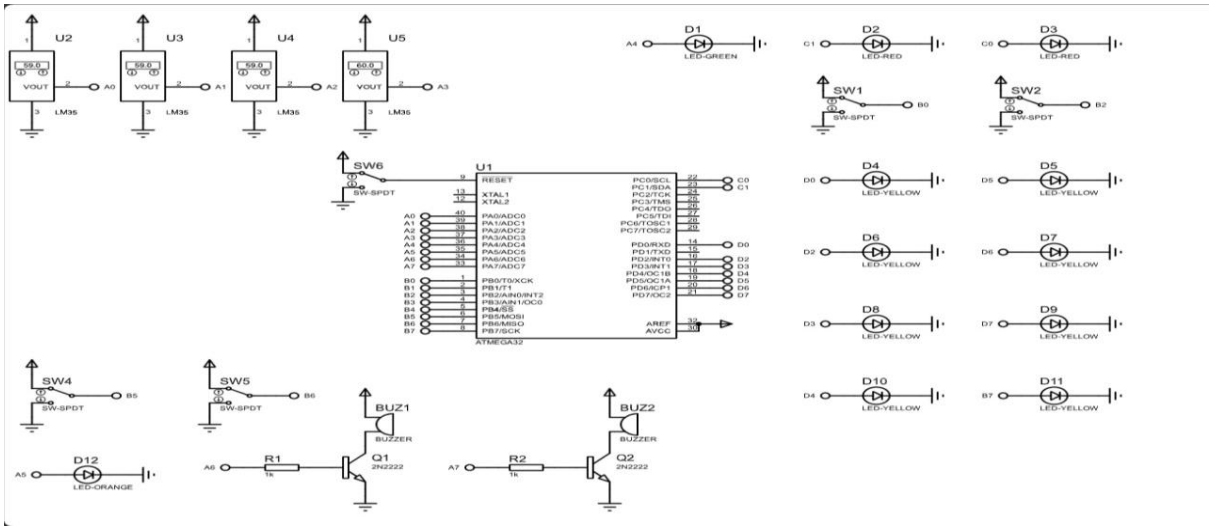


Figure 3.7: The main circuit design using Proteus software

The main circuit consist of four LM35 each one connected to zoon sensor inputs and labels as zoon1 to zoon4. The second componet is the push buttons UP1 for up and the number one, RT2 the right and number two, DN3 the down button and represent the number three, LF4 for the left button and the number four, MNU is the menu button and RST is the reset button. The direction keys used to nevegate or to select an item in the menue options. The healthy LED is a green LED that anly goes on, when all conditions are healthy, Z12RElAY zoon one and zoon two acuater relay this is the final output to release the fire

fighting liqued or gas to this area . Z34RELAY zoon three and zoon four acuater relay.

3.4 System Operation

the word ALRM ZON and FLT ZON then check the input analog heats element for any change also check the input keys, if one heat degree reach the maximum setting (60 centegrate) the alarm point the shows the alarm zoon with a number from one to four the alarm sounder operate and the zoons alarm red LED flashing, this situation stay as its as long as the alarm exist, when this tempreture goes down anly the flashing stop and the zoon lamp stay in on postion tell the user press reset key, when we press reset key the system check the alarm status clear and the sounder if the signal is normal. The same response if there is alarm at two not pair zoons like zoon one and three or zoon two and zoon four (the pair zoons is zoon one and two, zoon three and four).

Code c

```
/*
 * Project: Fire Alarm System (Dual Buzzer Edition)
 * MCU: ATmega32
 * Frequency: 8.000000 MHz
 * Description:
 * - Fire Buzzer on PA6 (Fast Beep)
 * - Fault Buzzer on PA7 (Slow Beep)
 * - No LCD
 * - Relays on PC0, PC1
 */

// ================== إعدادات النظام ==================
#define THRESHOLD_ALARM 60
#define THRESHOLD_FAULT 5   //
```

```
// ================== تعريف المنافذ ==================

// Outputs (Buzzers & LEDs)
#define BUZZER_FIRE   PORTA.F6 // PA6 -
#define BUZZER_FAULT  PORTA.F7 // PA7 -
#define LED_HEALTHY   PORTA.F4 // PA4 -
#define LED_ACT_SND   PORTA.F5 // PA5 -

// Outputs (Relays)
#define RELAY_Z34     PORTC.F0 // PC0 -
#define RELAY_Z12     PORTC.F1 // PC1 -

// Inputs (Switches - Active Low)
#define SW_UP1        PINB.F0  // PB0 -
#define SW_DN3        PINB.F2  // PB2
#define SW_ACK        PINB.F5  // PB5
#define SW_ACTR       PINB.F6  // PB6 -

// المتغيرات
short Fire_Alarm_Active = 0;
short Sensor_Fault_Active = 0;
short Buzzer_Muted = 0;

unsigned int temp_adc[4];
unsigned int temp_c[4];
short i;

// ================== الدوال المساعدة ==================

void System_Init() {
  // PORTA: PA0-PA3 (Input ADC), PA4-PA7 (Outputs)
  DDRA = 0xF0;
  PORTA = 0x10;
```

```
  // PORTB: Inputs (PB0-PB6), Output (PB7 for Zone4 Alarm)
  DDRB = 0x80;
  PORTB = 0x7F; //

  // PORTC: Outputs (Relays)
  DDRC = 0x03; // PC0, PC1 Output
  PORTC = 0x00;

  // PORTD: Outputs (Zone LEDs)
  DDRD = 0xFF;
  PORTD = 0x00;

  ADC_Init();
}

void Read_Temperatures() {
   for(i=0; i<4; i++) {
      temp_adc[i] = ADC_Read(i);
      // المعادلة: (ADC * 500) / 1024
      temp_c[i] = (unsigned long)temp_adc[i] * 500 / 1024;
   }
}

// ================== البرنامج الرئيسي ==================
void main() {
  System_Init();

  while(1) {
    Read_Temperatures();

    Fire_Alarm_Active = 0;
    Sensor_Fault_Active = 0;

    PORTD = 0x00;
    PORTB.F7 = 0;
```

```
  for(i=0; i<4; i++) {
    // فحص الحريق
    if(temp_c[i] >= THRESHOLD_ALARM) {
      Fire_Alarm_Active = 1;
      if (i < 3) PORTD |= (1 << (i + 5)); // Z1, Z2, Z3
      else PORTB.F7 = 1;              // Z4
    }
    // فحص العطل
    else if(temp_c[i] < THRESHOLD_FAULT) {
      Sensor_Fault_Active = 1;
      if (i == 0) PORTD.F0 = 1;
      else if (i == 1) PORTD.F2 = 1;
      else if (i == 2) PORTD.F3 = 1;
      else if (i == 3) PORTD.F4 = 1;
    }
  }

(Healthy) ---
  if (Fire_Alarm_Active || Sensor_Fault_Active) {
    LED_HEALTHY = 0
  } else {
    LED_HEALTHY = 1; /
  }

  if (Buzzer_Muted) {
    BUZZER_FIRE = 0;
    BUZZER_FAULT = 0;
    Delay_ms(100); /
  }
  else if (Fire_Alarm_Active) {
    BUZZER_FAULT = 0;

    BUZZER_FIRE = 1;
    LED_ACT_SND = 1;
    Delay_ms(100);       /
```

```
    BUZZER_FIRE = 0;      //
    LED_ACT_SND = 0;
    Delay_ms(100);        //
  }
  else if (Sensor_Fault_Active) {
    // ===
    //
    BUZZER_FIRE = 0; //

    BUZZER_FAULT = 1;     //
    Delay_ms(500);        //

    BUZZER_FAULT = 0;     //
    Delay_ms(500);        //
  }
  else {
    BUZZER_FIRE = 0;
    BUZZER_FAULT = 0;
    Delay_ms(100);
  }

  if((temp_c[0] >= THRESHOLD_ALARM) && (temp_c[1] >=
THRESHOLD_ALARM)) {
    RELAY_Z12 = 1;
  } else {
    RELAY_Z12 = 0;
  }

  if((temp_c[2] >= THRESHOLD_ALARM) && (temp_c[3] >=
THRESHOLD_ALARM)) {
    RELAY_Z34 = 1;
  } else {
    RELAY_Z34 = 0;
  }
```

```
  // --- 5.

  // تشغيل يدوي Z12
  if (SW_UP1 == 0) RELAY_Z12 = 1;

  // تشغيل يدوي Z34
  if (SW_DN3 == 0) RELAY_Z34 = 1;

  // زر كتم الصوت (ACK)
  if (SW_ACK == 0) {
     Delay_ms(20); // Debounce
     if (SW_ACK == 0) Buzzer_Muted = 1;
  }

  // زر إعادة تفعيل الصوت (Reset)
  if (SW_ACTR == 0) {
     Delay_ms(20);
     if (SW_ACTR == 0) Buzzer_Muted = 0;
  }
 }
                                                    }
```

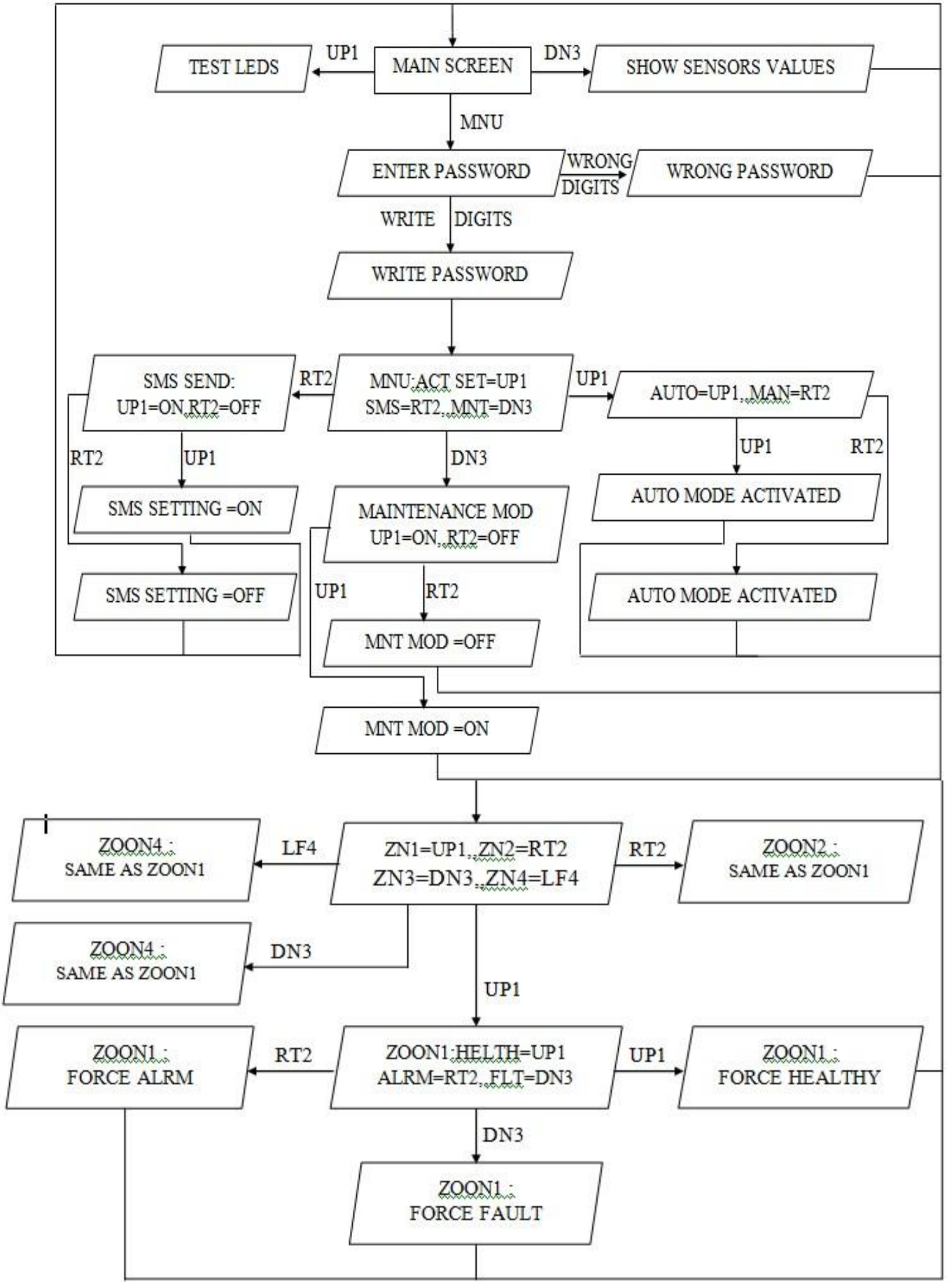
assmbely_code

```
.include "m32def.inc"
; ================= REGISTER DEFINITIONS =================
.def tempL = r16
.def tempH = r17
.def zone  = r18
.def flags = r19
.def d1    = r20
.def d2    = r21
.equ THRESHOLD_ALARM = 60
.equ THRESHOLD_FAULT = 5
.org 0x0000
```

```
rjmp RESET
RESET:
; Stack
ldi r16, high(RAMEND)
out SPH, r16
ldi r16, low(RAMEND)
out SPL, r16
; PORTA
ldi r16, 0b11110000
out DDRA, r16
ldi r16, 0b00010000
out PORTA, r16
   ; PORTB
   ldi r16, 0b10000000
   out DDRB, r16
   ldi r16, 0b01111111
   out PORTB, r16
   ; PORTC
   ldi r16, 0b00000011
   out DDRC, r16
   clr r16
   out PORTC, r16
   ; PORTD
   ldi r16, 0xFF
   out DDRD, r16
   clr r16
   out PORTD, r16
   ; ADC
   ldi r16, (1<<REFS0)
   out ADMUX, r16
   ldi r16, (1<<ADEN)|(1<<ADPS2)|(1<<ADPS1)
   out ADCSRA, r16

   clr flags
```

```
MAIN:
   clr zone
   clr flags
ZONE_LOOP:
cpi zone, 4
breq MAIN
; Select ADC channel
ldi r16, (1<<REFS0)
or  r16, zone
out ADMUX, r16
sbi ADCSRA, ADSC
WAIT_ADC:
sbis ADCSRA, ADIF
rjmp WAIT_ADC
sbi ADCSRA, ADIF
in tempL, ADCL
in tempH, ADCH
; Fake temperature using high byte
mov r16, tempH
cpi r16, THRESHOLD_ALARM
brlo NEXT_ZONE
sbr flags, (1<<0)
NEXT_ZONE:
inc zone
                                    rjmp ZONE_LOOP
```


TEST LEDS
UP1
MAIN SCREEN
DN3
SHOW SENSORS VALUES
MNU
ENTER PASSWORD
WRONG DIGITS
WRONG PASSWORD
WRITE DIGITS
WRITE PASSWORD
SMS SEND: UP1=ON,RT2=OFF
RT2
MNU:ACT SET=UP1 SMS=RT2, MNT=DN3
UP1
AUTO=UP1, MAN=RT2
RT2
UP1
DN3
UP1
RT2
SMS SETTING =ON
MAINTENANCE MOD UP1=ON, RT2=OFF
AUTO MODE ACTIVATED
SMS SETTING =OFF
UP1
RT2
AUTO MODE ACTIVATED
MNT MOD =OFF
MNT MOD =ON
ZOON4 : SAME AS ZOON1
LF4
ZN1=UP1, ZN2=RT2 ZN3=DN3, ZN4=LF4
RT2
ZOON2 : SAME AS ZOON1
ZOON4 : SAME AS ZOON1
DN3
UP1
ZOON1 : FORCE ALRM
RT2
ZOON1:HELTH=UP1 ALRM=RT2, FLT=DN3
UP1
ZOON1 : FORCE HEALTHY
DN3
ZOON1 : FORCE FAULT

Chapter Five

Conclusion and Recommendations

5.1 Conclusion

In this study a fire fighting system is build up using Atmega32 microcontroller. The system is tested and performs successfully and achieved all required operation conditions. In addition to that there is interactive with the system through SMS notification.

5.2 Recommendations

- Upgrade the system to be remote controlled.
- Design farther types of fire detectors.
- System development and adaptation for use as integrated systems.

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